Specification of Feature Isolation

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1 Basic Information and Results

Feature isolation is based on the following information:

- A specification of a software product line including the names of its independent features, the feature categories to be considered, the maximum rank of interaction between any kind of and- and or-features, and optionally explicitly dependent features plus available systems or missing systems all of which resulting in a set with the identifiers of available systems. This specification is denoted by *SPL*. An identifier (*ID* for short) is a natural number with a lower bound of 1.
- For each feature there exists a description that specifies in which systems it occurs and in which it does not plus an associated unique feature *ID*. Such a feature description is called *FD*, a feature *ID* is called *FID*.
- Each system has a description that specifies the features it consists of plus an associated unique system *ID*. Such a system description is called *SD*, a system *ID* is called *SID*.

Evaluation. To describe the isolation of a feature f_i based on its *FID* some operations are introduced.

- Given is a system description *SD*.
- *SD* can be flattened by replacing all each of its elements by the corresponding set of *FID*s and applying either set intersection, , or set union, , to them.
- Flattening *SD* with gives *SD*.
- Flattening *SD* with gives *SD*.

The isolation of a feature f_i specified by its *FID* involves the following steps:

1. I is the set with *SID*s of all systems that include FID_i.
2. U is the set with *SID*s of all systems that do *not* include FID_i.
3. $I\,U$ gives the result of feature isolation which is denoted by $\overline{D}$. $\overline{D}$ must not be equal to as this would indicate a fundamental error during the preceding steps. $\overline{D}$ can include exactly one element which then must be the FID_i. If $\overline{D}$ includes more than one element, one of them must be FID_i.

Minimal set differences Given are the following definitions:

1. $D\ \ IU$
2. $\overline{D}\ \ I\,U$
3. $I_S\ \ I$
4. $U_S\ \ U$
5. $D_S\ \ D\ \ I_S\ \ I\ \ U_S\ \ U$

Then any D that evaluates to $\overline{D}$ if there is no proper subset of D which also gives $\overline{D}$ is *minimal*.

There exists at least one subset of D which gives $\overline{D}$, namely D. There can be more than one proper subset of D which also gives $\overline{D}$. The set that includes all different subsets of D that evaluate to $\overline{D}$, is called M. $M\ \ 1$.

THE PRECEDING TWO PARAGRAPHS SHOULD BE MORE FORMALIZED.

The following examples illustrate the meaning of *minimal* in a set difference (the elements are *FID*s. Given are the systems
$SD_1\ \ 1, 2, 3,$
$SD_2\ \ 2, 3, 4,$
$SD_3\ \ 3, 4, 5,$
$SD_4\ \ 4, 5, 6,$
$SD_5\ \ 2, 7$ and
$SD_6\ \ 1, 8.$

With respect to FID_2, the following set intersections I are minimal as they all isolate FID_2.
$I_1\ \ SD_1\ \ SD_2$
$I_2\ \ SD_1\ \ SD_5$
$I_3\ \ SD_2\ \ SD_5$

$I_4\ \ SD_1\ \ SD_2\ \ SD_5$ is not minimal because I_1, I_2 and I_3 successfully isolate FID_2 but contain less operands. It is important to note that only evaluating I can give the desired result. Furthermore, I cannot be omitted.

With respect to FID_2 the set difference $D_1\ \ SD_1SD_3\ \ SD_6$ is minimal as it isolates FID_2.

$D_2\ \ SD_1\ \ SD_2SD_3$ is not minimal, as I already isolates FID_2. $D_3\ \ SD_1SD_3\ \ SD_4\ \ SD_6$ is also not minimal, as $SD_3\ \ SD_6$ in U already removes all unwanted elements of I. It has to be emphasized, that —different from I— U can be omitted.

2 Functions, Datatypes and Variables

- The datatypes `feature_ids_t` and `system_ids_t` represent the set types for *FID*s and *SID*s.
- The member variable `spl_specification_t::m_available_systems` which is of type `system_ids_t` contains the *SID*s of the actually available systems.
- `difference_t` is the type that represents a single set difference. It defines the data members `I` and `U`, both of type `system_ids_t`.They represent the operands I and U of a set difference that isolates a feature f_i.
- `minimal_differences_t` represents M. It defines the data members `features` which is of type `feature_ids_t` and represents D